📦 Unit 9: Volume Practice — ANSWER KEY

Cylinders · Cones · Spheres · Composite Solids · NC.8.G.9

Name: _____ Date: _____ Period: _____

Cylinder: $V = \pi r^2 h$ **Cone: $V = \frac{1}{3}\pi r^2 h$** **Sphere: $V = \frac{4}{3}\pi r^3$**

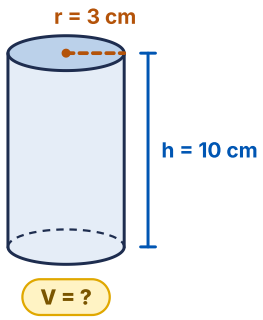
π ≈ 3.14 · Always use cubic units · Remember: radius = diameter ÷ 2

💡 **Tip:** A cone is exactly ⅓ of a cylinder with the same dimensions. When you see "diameter," divide by 2 to get radius first!

🥫 Cylinder Problems

Problem 1 **Score 3**

A cylinder has radius 3 cm and height 10 cm. What is the volume? (Use π ≈ 3.14)

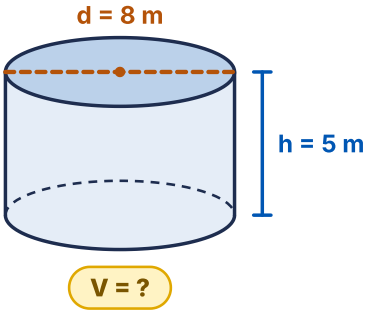


- ✓ **282.6 cm³**
- ○ 94.2 cm³
- ○ 188.4 cm³
- ○ 1130.4 cm³

Solution: $V = \pi(3)^2(10) = \pi(9)(10) = 90\pi \approx 282.6$ cm³

Problem 2 **Score 3**

A cylinder has diameter 8 m and height 5 m. What is the volume?

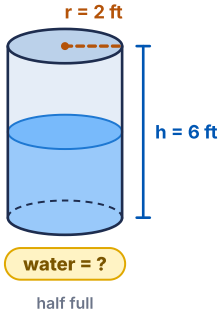


- ✓ **80π m³**
- ○ 160π m³
- ○ 320π m³
- ○ 40π m³

Solution: Diameter = 8, so r = 4. $V = \pi(4)^2(5) = \pi(16)(5) = 80\pi \approx 251.2$ m³

Problem 3 **Score 4**

A cylindrical water tank (r = 2 ft, h = 6 ft) is half full. How much water is inside?



- ✓ **12π ft³**
- ○ 24π ft³
- ○ 6π ft³
- ○ 8π ft³

Solution: Full: $V = \pi(2)^2(6) = 24\pi$. Half full: $24\pi \div 2 = 12\pi$ ft³

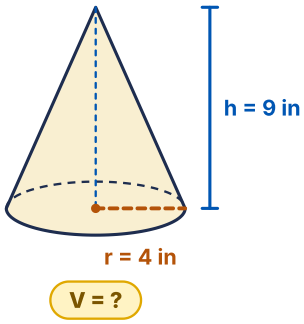
🍦 Cone Problems

Problem 4 Score 3

A cone has radius 4 in and height 9 in. What is the volume?

- ○ 64π in³
- ✔ **48π in³**
- ○ 452.16 in³
- ○ 36π in³

Solution: $V = \frac{1}{3}\pi(4)^2(9) = \frac{1}{3}\pi(16)(9) = \frac{1}{3}(144\pi) = 48\pi \approx 150.72$ in³

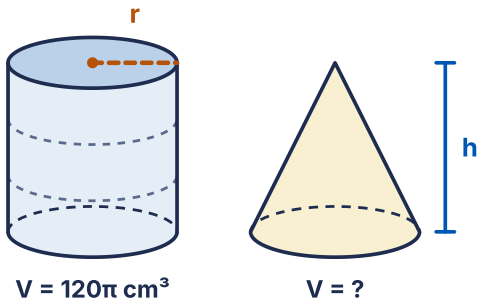


Problem 5 Score 4

A cone and cylinder have the same radius and height. The cylinder's volume is 120π cm³. What is the cone's volume?

- ✔ **40π cm³**
- ○ 60π cm³
- ○ 360π cm³
- ○ 80π cm³

Solution: A cone is ⅓ of a cylinder! 120π ÷ 3 = 40π cm³



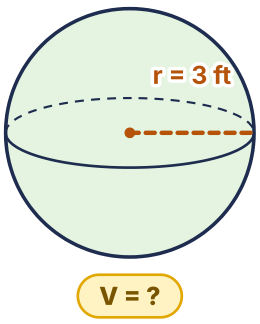
🏀 Sphere Problems

Problem 6 Score 3

A sphere has radius 3 ft. What is the volume?

- ✔ **36π ft³**
- ○ 27π ft³
- ○ 84.78 ft³
- ○ 108π ft³

Solution: $V = \frac{4}{3}\pi(3)^3 = \frac{4}{3}\pi(27) = 36\pi \approx 113.0$ ft³

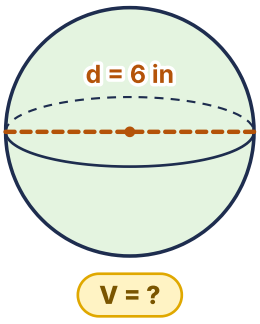


Problem 7 Score 4

A sphere has diameter 6 in. What is the volume?

- ✔ **36π in³**
- ○ 288π in³
- ○ 12π in³
- ○ 27π in³

Solution: Diameter = 6, so r = 3. $V = \frac{4}{3}\pi(3)^3 = \frac{4}{3}\pi(27) = 36\pi \approx 113.0$ in³



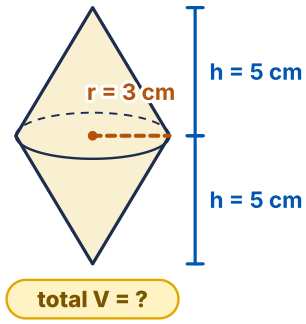
📦 Mixed & Composite

Problem 8 Score 5

Two identical cones are placed base-to-base. Each cone has r = 3 cm and h = 5 cm. What is the total volume?

- ✔ **30π cm³**
- ○ 60π cm³
- ○ 15π cm³
- ○ 90π cm³

Solution: One cone: V = ⅓π(3)²(5) = 15π. Two cones: 2(15π) = 30π cm³

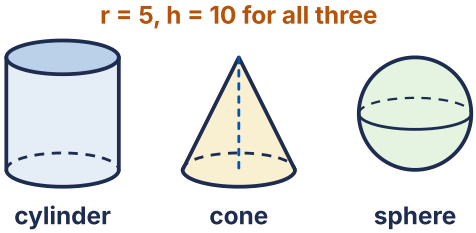


Problem 9 Score 5

Which shape has the greatest volume? r = 5, h = 10 for all.

- ✔ **Cylinder**
- ○ Cone
- ○ Sphere
- ○ They're all equal

Solution: Cylinder: π(25)(10) = 250π. Cone: ⅓(250π) ≈ 83.3π. Sphere: ⁴⁄₃π(125) ≈ 166.7π. Cylinder wins!

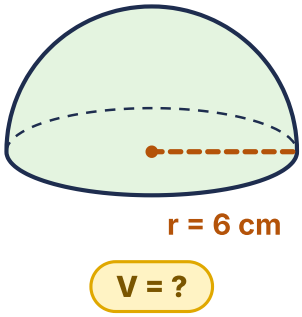


Problem 10 Score 5

A hemisphere (half sphere) has radius 6 cm. What is its volume?

- ✔ **144π cm³**
- ○ 288π cm³
- ○ 72π cm³
- ○ 216π cm³

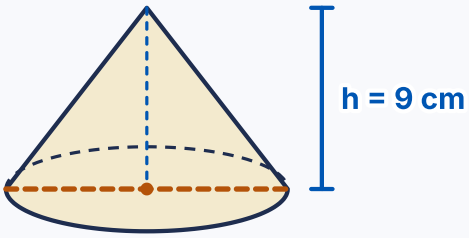
Solution: Full sphere: ⁴⁄₃π(6)³ = ⁴⁄₃π(216) = 288π. Hemisphere = half: 288π ÷ 2 = 144π cm³



📝 Worked Examples

Study these step-by-step solutions before attempting the problems above.

Volume of a cone: diameter = 14 cm, height = 9 cm



V = ?

diameter 14 → radius 7

Step 1: Identify r — diameter = 14, so r = 7

$r = 7,\ h = 9$

Step 2: Write the formula

$V = \frac{1}{3}\pi r^2 h$

Step 3: Substitute

$V = \frac{1}{3}\pi(7^2)(9) = \frac{1}{3}\pi(49)(9)$

Step 4: Multiply inside

$V = \frac{1}{3}(441\pi)$

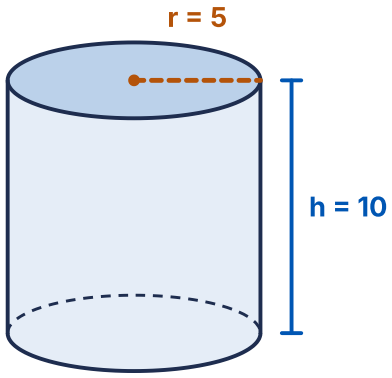
Step 5: Divide by 3

$V = 147\pi \approx 461.6\ \text{cm}^3$ ✓

🔍 Common Mistakes — Can You Spot the Error?

Each problem has a mistake. Identify what went wrong.

Cylinder: r = 5, h = 10

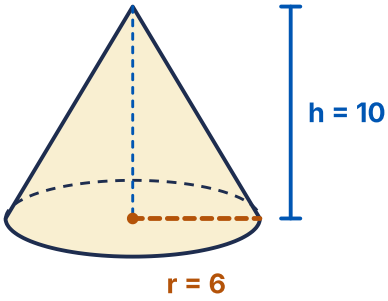


✗ **Student:** $V = \pi(5)(10) = 50\pi$

✓ **Fix:** Forgot to SQUARE the radius! $V = \pi(5^2)(10) = \pi(25)(10) = 250\pi$

💡 *The formula is $\pi r^2 h$ — that ^{2} is critical!*

Cone: r = 6, h = 10

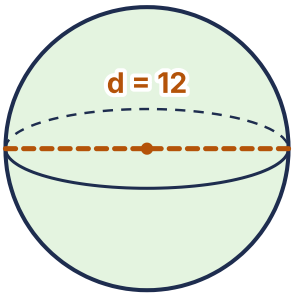


✗ **Student:** $V = \pi(6^2)(10) = 360\pi$

✓ **Fix:** That's the CYLINDER formula! Cones need ⅓. $V = \frac{1}{3}\pi(36)(10) = 120\pi$

💡 *A cone is always ⅓ of the matching cylinder.*

Sphere: diameter = 12



the 12 goes all the way across

✗ **Student:** $V = \frac{4}{3}\pi(12)^3 = \frac{4}{3}\pi(1728) = 2304\pi$

✓ **Fix:** Used DIAMETER instead of radius! r = 6, not 12. $V = \frac{4}{3}\pi(6)^3 = \frac{4}{3}\pi(216) = 288\pi$

💡 *Always check: did the problem give diameter or radius? Diameter ÷ 2 = radius.*